

THE PROBLEM: THE ECO-PARALYSIS PARADOX

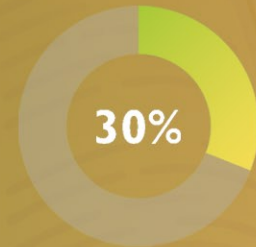
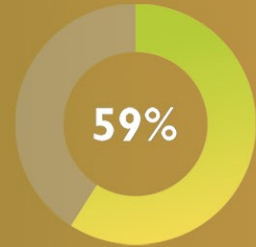
Current schooling builds awareness but fails to trigger action, stalling **SDG 4**. Global data reveals a critical fracture in the education system:



Eco-Anxiety (The Feeling): 59% of youth are "extremely worried" about climate, living in a state of psychological stress (*The Lancet*, 2021).

Competence (The Reality): Despite anxiety, only 30% could actually act, due to a lack of education and integration in curricula (*UNESCO*, 2021).

Problem: A 29% gap (Paralysis) exists between those fearing the future and those with tools to change it. No personalization means fear, not skill.



SOLUTION: HYPER-PERSONALIZED EDUCATION

To bridge the gap, we use *AI* for Hyper-Personalized Education. Data confirms that smart adaptation maximizes engagement and learning impact:

Performance (*RAND*): Personalized learning boosts student performance by **10 percentage points** (Math/Reading, *Chart 1*), significantly improving the average student's standing (+30% over the standard average).

Completion (*Engageli*): AI systems improve motivation and outcomes by up to **30%** (in 75% of students) through adaptive learning pathways.

Retention (*Bower & Clark*): Emotional storytelling increases information recall by **80%**.



DATA SCIENCE: **LIFECYCLE & TEAM**

"Green Whisper" applies these principles technically:

Input:

Automated collection of local geospatial data
(e.g. pollution, biodiversity)

Processing:

A Generative DL model turns cold data into empathetic visual stories
("Your neighborhood in 2035"), curated by **Data Scientists**
(Analysis) and **ML Engineers** (Generative Models).

Output:

Hyper-realistic educational experience leveraging the retention
boost cited in Slide 2

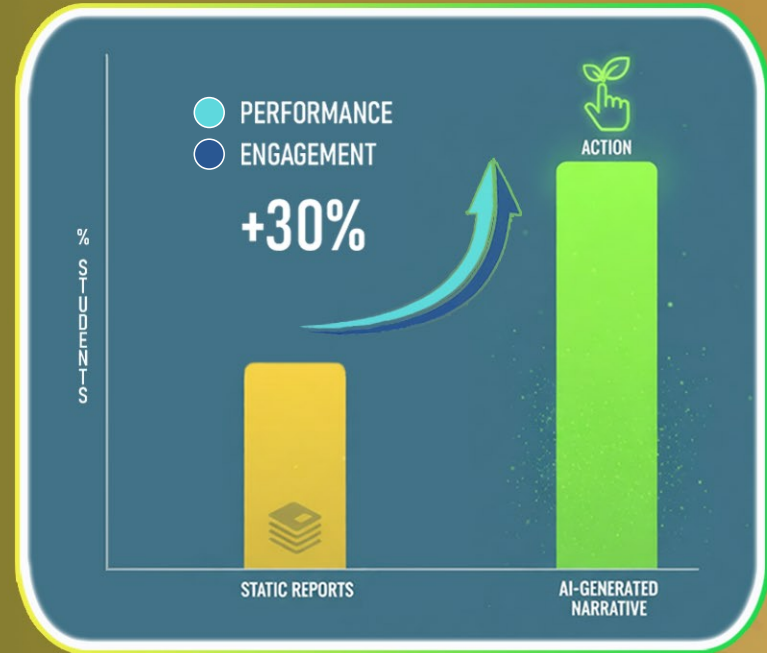
THE CALCULATION: IMPACT PROJECTION

Applying AI efficacy factors (Slide 2) to the competence deficit (Slide 1), we project the social change:

State A (Today): Only the 31% have active competence (Baseline).

Boost AI: By applying performance gains (+30%, *RAND*) and conservative engagement factors (+30%, *Engageli*), we estimate a direct increase in practical awareness.

State B (Target): The active student population would rise to a range of about 61%. A double-digit increase mobilizing millions of "environmental sentinels," concretely achieving I'SDG 4.7.



SOURCES

PROBLEM & GAP: Marks, E., et al. (2021). Young People's Voices on Climate Anxiety. The Lancet Planetary Health. DOI: [10.1016/S2542-5196\(21\)00278-3](https://doi.org/10.1016/S2542-5196(21)00278-3) | UNESCO (2021). Learn for our planet. [Link](#)

SOLUTION DATA: Pane, J.F., et al. (2015). Continued Progress: Promising Evidence on Personalized Learning. RAND Corporation. [Link](#) | Bower, G.H., & Clark, M.C. (1969). Narrative stories as mediators for serial learning. Psychonomic Science. DOI: [10.3758/BF03332778](https://doi.org/10.3758/BF03332778)

AI STATISTICS: Engageli (2025). AI in Education Statistics. [Link](#)